

Core Branches: NLP, Computer Vision & Predictive ML

Natural Language Processing (NLP)

NLP enables machines to read, understand, and generate human language. It is the branch behind chatbots, search engines, translation, sentiment analysis, and large language models.

- Core tasks: tokenisation, named entity recognition, machine translation, summarisation, question answering.
- Breakthrough: the Transformer architecture (2017) enabled context-aware language understanding at scale.
- Real-world examples: Google Search, ChatGPT, email autocomplete, spam filters.

NLP bridges the gap between human communication and machine processing.

Computer Vision (CV)

Computer Vision enables machines to interpret and understand visual information from the world — images, video, and real-time camera feeds.

- Core tasks: image classification, object detection, image segmentation, facial recognition, optical character recognition (OCR).
- Key architecture: Convolutional Neural Networks (CNNs) — AlexNet (2012) was the watershed moment.
- Real-world examples: Face ID, self-driving car perception, medical image diagnosis (radiology), quality control in manufacturing.

Computer Vision gives machines 'eyes' — turning pixels into actionable understanding.

Predictive Machine Learning (Predictive ML)

Predictive ML uses historical data to forecast future outcomes. It is the workhorse of business intelligence, finance, and healthcare analytics.

- Core tasks: regression (predict a number), classification (predict a category), time-series forecasting.
- Key algorithms: Linear Regression, Random Forests, Gradient Boosting (XGBoost), LSTM networks.
- Real-world examples: credit scoring, demand forecasting, disease risk prediction, churn prediction.

Comparison of the Three Branches

Branch	Input Type	Output Type	Key Example
NLP	Text / Speech	Text / Labels	ChatGPT, Google Translate
Computer Vision	Images / Video	Labels / Boxes	Face ID, Radiology AI
Predictive ML	Tabular / Time-series	Numbers / Labels	Credit scores, forecasts

Key Takeaways

- NLP: language in, language or labels out — powers chatbots and search engines.
- Computer Vision: images/video in, understanding out — powers Face ID and autonomous vehicles.
- Predictive ML: historical data in, forecasts out — powers credit and demand systems.

Quick-Revision Questions

- Name two real-world applications of each of the three AI branches.
- What architecture powers modern NLP? What architecture powers Computer Vision?
- How does Predictive ML differ from the other two branches in terms of input data?